Examples of Discrete Random Variables and their probability distribution

Bernoulli Random Variable $X \sim Bern(p)$

Definition

We say a random variable X has a Bernoulli distribution if it only takes two values, 1 or 0.
Bernoulli is used to model one trial when there is only two possible outcomes; success or failure, where $P(success) = p$.
When $X \sim Bern(p)$, then

$$p_X(x) = \begin{cases} 1 - p & x = 0 \\ p & x = 1 \end{cases}$$

and

$$E[X] = p$$
$$Var(X) = p(1 - p)$$

Binomial Distribution $X \sim Bin(n, p)$

It is used to model number of successes in n independent Bernoulli trials. Or it is used to model number of successes in n independent trial, where probability success in each is p. or number of heads in n independent toss of a coin where probability of head is p.

Definition

We say $X \sim Bin(n, p)$, then
X: = number of success in n independent trail
X can take values, $0, 1, 2, \cdots, n$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}; \text{ where } k = 1, 2, \cdots, n$$
$$E[X] = np$$
$$Var(X) = np(1-p)$$

Eg 1. Suppose that a box contains 3 blue blocks and 2 red blocks. You pick four blocks with replacement.
Define success as "picking a blue block".
X = number of successes.
What is the expected number of blue blocks (i.e. the mean)?
What are the variance and standard deviation?

It's draw with replacement; meaning trials are independent and p(success) is fixed.

In each draw either the block is blue or not-blue

$$p(\text{success}) = P(\text{blue block}) = \frac{3}{5}$$

$$X \sim \text{Bin}\left(n = 4,\ p = \frac{3}{5}\right)$$

$$E[X] = np = 4 \cdot \frac{3}{5} = \frac{12}{5}, \quad \text{Var}(X) = np(1-p) = \frac{24}{25}$$

When can we use Binomial Distribution? When all the followings are true.

1. The number of trials, n, is fixed.
2. Each trial has two possible outcomes: "success" and "failure".
3. The probability of "success", p, is the same from trial to trial.
4. The trials are independent.
5. X = number of "successes" in n independent trials.

Note: Read this only if you are interested in modelling, not for current course. In real world, sometimes not all the above is true, for example if trials are dependent or probability of success is not fixed. For example we are interested in number of white balls in n draw, but we are drawing without replacement. Then we can still use Binomial for approximation if the sample size is much smaller than the whole population size.

Using CDF table for Calculating probabilities

Eg. Find $P(4 \leq X \leq 7)$ where $X \sim Bin(n = 15, p = 0.2)$.

$$\begin{aligned} P(4 \leq X \leq 7) &= F(7) - F(4-) = F(7) - F(3) \\ &= B(7, n = 15, p = 0.2) - B(3, n = 15, p = 0.2) \end{aligned}$$

In the Book it use B for Binomial distribution

From the table in next page

$$P(4 \leq X \leq 7) = 0.996 - 0.648$$
$$= 0.348$$

Note: This is a only one of the tables. The book has more values for n and p.

Note: A table entry = 0 does not indicate an impossibility. All table entries are actually positive, but rounded to three significant digits appear as 0.000.

c. $n = 15$

		p														
		0.01	**0.05**	**0.10**	**0.20**	**0.25**	**0.30**	**0.40**	**0.50**	**0.60**	**0.70**	**0.75**	**0.80**	**0.90**	**0.95**	**0.99**
	0	.860	.463	.206	.035	.013	.005	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.990	.829	.549	.167	.080	.035	.005	.000	.000	.000	.000	.000	.000	.000	.000
	2	1.000	.964	.816	.398	.236	.127	.027	.004	.000	.000	.000	.000	.000	.000	.000
	3	1.000	.995	.944	.648	.461	.297	.091	.018	.002	.000	.000	.000	.000	.000	.000
	4	1.000	.999	.987	.836	.686	.515	.217	.059	.009	.001	.000	.000	.000	.000	.000
	5	1.000	1.000	.998	.939	.852	.722	.403	.151	.034	.004	.001	.000	.000	.000	.000
	6	1.000	1.000	1.000	.982	.943	.869	.610	.304	.095	.015	.004	.001	.000	.000	.000
x	**7**	1.000	1.000	1.000	.996	.983	.950	.787	.500	.213	.050	.017	.004	.000	.000	.000
	8	1.000	1.000	1.000	.999	.996	.985	.905	.696	.390	.131	.057	.018	.000	.000	.000
	9	1.000	1.000	1.000	1.000	.999	.996	.966	.849	.597	.278	.148	.061	.002	.000	.000
	10	1.000	1.000	1.000	1.000	1.000	.999	.991	.941	.783	.485	.314	.164	.013	.001	.000
	11	1.000	1.000	1.000	1.000	1.000	1.000	.998	.982	.909	.703	.539	.352	.056	.005	.000
	12	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.996	.973	.873	.764	.602	.184	.036	.000
	13	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.965	.920	.833	.451	.171	.010
	14	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.987	.965	.794	.537	.140

(continued)

Eg. 3 An salesman thinks that the probability of making a sale is 0.30. If he talks to five customers on a particular day, what is the probability that he will make exactly 2 sales? (Assume independence.)

sale = success

$p = P(\text{success}) = 0.3$

$n = 5$

$X :=$ number of sales in interaction with 5 customers

$$P(X=2) = \binom{5}{2} p^2 (1-p)^3 = \binom{5}{2} (0.3)^2 (0.7)^3$$

$$= 10\,(0.09)\,(0.343) = 0.3087$$